Published in final edited form as:
Adv Mater Technol. 2025 January 08; 10(1): . doi:10.1002/admt.202570002.

Design of Modular, 3D-Printed Millifluidic Mixers to Enable Sequential NanoPrecipitation (SNaP) for the Tunable Synthesis of Drug-Loaded Nanoparticles and Microparticles

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Abstract

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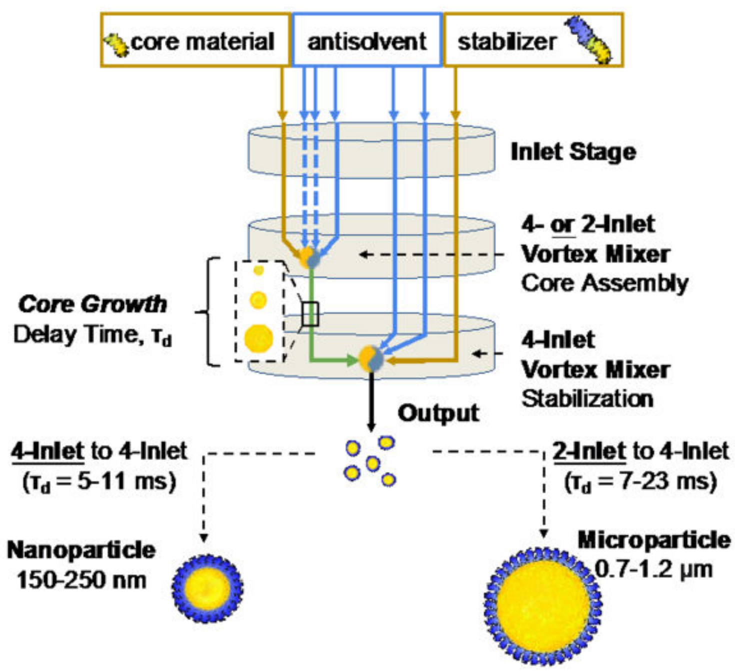
Sequential NanoPrecipitation (SNaP) is a nascent controlled precipitation process for the tunable formation of polymeric particles for drug delivery and bioimaging. While SNaP relies on the same self-assembly principles as one-step Flash NanoPrecipitation, SNaP is a two-step assembly process in which the particle core is formed during a first mixing step followed by particle stabilization in a second mixing step. Decoupling the particle assembly steps improves control over the particle structure and, as we demonstrate for the first time, expands the attainable particle size range to include microparticles. Current SNaP experimental set-ups use commercial millifluidic mixers connected in series that suffer from several drawbacks including the inability to access short inter-mixer delay times. Here, we develop a robust 3D-printed, modular mixer design that enables access to short delay times (< 25 ms) not previously accessible. We prove empirically for the first time that the inter-mixer delay time is a key parameter for particle size control and that the nanoparticle size scales with delay time in agreement with Smoluchowski's model of diffusion-limited growth. We demonstrate the formation of polymeric particles ranging in size from 160 nm to 1.2 μm. Finally, we establish the versatility and applicability of our mixer design by encapsulating fluorophores and therapeutics into particles for the first time via SNaP.

Graphical Abstract

We develop a new modular sequential mixer platform design to enable Sequential NanoPrecipitation (SNaP), an emerging controlled precipitation process for the formation of particles for drug delivery and bioimaging applications. We show that the inter-mixer delay time is a key process parameter for controlling particle size. For the first time, we demonstrate the encapsulation of therapeutics into nanoparticles and microparticles via SNaP.



Keywords

sequential nanoprecipitation; flash nanoprecipitation; nanoparticle; microparticle; 3D printing; millifluidic mixer design; controlled drug delivery

1. Introduction

Polymeric drug delivery vehicles have attracted a great deal of interest over the last three decades due to their ability to controllably deliver small molecule and biologics-based therapeutics.[1–5] Polymeric drug delivery vehicles have been shown to improve drug bioavailability, exhibit sustained drug release, and both passively and actively target tissues and cells.[6–12] These capabilities contribute to a decrease in off-target toxicities and an improvement in drug efficacy compared to free drug.[13, 14] Moreover, polymeric vehicles can be coupled with imaging agents to create medical contrast agents for in vitro and in vivo imaging.[15–17] Despite these advantages, the commercial translation of polymeric vehicles has been limited in part by challenges related to the scaling of their synthesis from the lab scale to the industrial scale.[18] Few processes can be scaled while maintaining consistent vehicle physicochemical properties. One process that has been successfully scaled is Flash Nanoprecipitation (FNP), which is the technical foundation for the new Sequential NanoPrecipitation (SNaP) process.[19]

FNP is a one-step, controlled precipitation process for the formation of polymeric core-shell nanoparticles (NPs) from 50 to 400 nm in size with independent control over the size, drug loading and surface functionality.[20, 21] In FNP, hydrophobic molecules (i.e. therapeutic drugs, dyes, etc.) and amphiphilic block-copolymers (BCP) are dissolved in an organic solvent stream, which is rapidly mixed against aqueous anti-solvent streams using millifluidic mixers.[22, 23] To achieve the required turbulent micromixing, the FNP process uses confined impinging jet mixers (CIJ)[26] or multi-inlet vortex mixers (MIVM) operating at high Reynolds numbers (Re > 2000).[27] These mixers generate a micromixing environment which leads to a rapid change in solvent quality on the 1–3 millisecond timescale and a homogenous supersaturation of the hydrophobic components prior to NP assembly.[24] This is followed by the rapid nucleation and growth of the hydrophobic core and the adsorption of the block copolymer to stabilize the NPs, which occurs on the 10–50 millisecond timescale.[21, 25] The resulting polymeric core-shell NPs have cores containing hydrophobic materials and therapeutics, and a corona comprised of the hydrophilic block of the BCP.[20, 26]

Despite FNP's success and wide adoption in academia and industry, FNP has four major limitations: (1) FNP cannot synthesize stable NPs above 400 nm in diameter; (2) The maximal drug loading attainable via FNP is around 70 wt% drug in NPs; (3) Due to entrainment of stabilizing polymers at high organic stream concentrations, the maximum solids content at the outlet is approximately 0.1 wt%;[21] And (4) to achieve the necessary nucleation rates, only very hydrophobic molecules (LogP > 5)[28] or those compatible with hydrophobic ion pairing can be encapsulated.[29,30] These limitations are a result of the core formation and stabilization occurring simultaneously in the single-step process.

While parameters such as solvent quality[31] and mixing rates[27,32] have been shown to affect particle size for single-step assembly of block copolymer systems, there has yet to be a strategy for controlling the growth of hydrophobic cores prior to stabilization. Recently, two new assembly processes have been developed that have the potential to overcome these limitations while retaining all the advantages of FNP: quench FNP (qFNP) [33] and Sequential NanoPrecipitation (SNaP).[34] In both processes, the core formation and NP stabilization are decoupled and occur sequentially with the core formation initiated first followed by stabilization. These evolutions of FNP rely on the same controlled assembly principles as FNP and use turbulent micromixing to ensure homogenous material distribution prior to assembly events.

In the SNaP process, there are two sequential mixing steps. In the first mixing step, the dissolved hydrophobic core components are rapidly mixed against an aqueous anti-solvent, which initiates the nucleation and growth of the nanoparticles core. In the second mixing step, the stabilizing agent is added, which adsorbs onto the growing core surface, arresting the growth and stabilizing the NPs (Figure 1A). The stabilizer addition step must occur within a few milliseconds of core growth initiation to control the final particle size distribution and prevent uncontrolled aggregation and precipitation. Current SNaP set-ups connect two traditional FNP mixers in series (CIJ/MIVM).[34] The outlet of the first mixer flows directly into an inlet of the second mixer. The tube length and injection flow rates control the delay time (T_d) between the two mixing steps. Current SNaP approaches use two configurations. The first configuration is a two-inlet CIJ mixer connected to a four-inlet MIVM, and the second configuration is a four-inlet MIVM mixer connected to another four-inlet MIVM mixer. In both cases, standard tubing is used to connect the mixers.[34]

Although these two mixing configurations have successfully been used for the formation of polymeric nanoparticles with a size ranging between 200 – 300 nm, their use remains limited.[34] This is due in part to the high cost of the CIJ and MIVM mixers, which are typically purchased from private manufacturers. Moreover, researchers are restricted to those commercially available CIJ and MIVM designs and dimensions. Depending on the synthesis requirements, modifications to the mixer geometries may be desired. However, the most significant limitation is the need for external tubing to connect the mixers in the currently available SNaP mixer set-ups. The length of the tubing controls the T_d's achievable between the mixing steps, which, as we will demonstrate herein, is a critical process parameter. With external tubing, it is challenging to connect the mixers precisely. Inconsistencies between SNaP mixer set-ups will be observed if the T_d between the two steps is not precisely controlled. Most importantly, achieving the short T_d's required for small NP formation (< 200 nm) is not feasible with external tubing and adapters. The shortest tube lengths attainable, essentially the length of two connection adapters, are still too long to attain the required T_d's even when operating at the highest flow rates achievable with syringe pumps. Finally, connecting the mixers using external tubing runs the risk of clogging due to kinked or twisted tubing.

In this work, we present a 3D-printed approach for rapidly prototyping millifluidic mixers for SNaP that solves the challenges of cost and accessible T_d's. Recent advancements in 3D printing now allow for sufficient printing resolution to print the narrow channels required for milli-fluidic mixers.[36–38]Through this prototyping, we developed a new mixer design that avoids the use of external tubing between the mixing steps, allowing us to access much shorter T_d's than previous designs and prevent leaks. The novel design combines two mixers into a single set-up. This was achieved with a "stacked" mixer design, where each step takes place in an interchangeable 3D printed stage, placed one on top of the other (Figure 1D). The first stage is the inlet stage, in which the solutions are injected, followed by an initial mixing stage with either a two- or a four-inlet vortex mixer. The output of this first mixing stage becomes an inlet of the second mixing stage, which consists of a four-inlet vortex mixer. Because of the well-controlled channel length between the two mixers, this stacked design provides better control over the particle synthesis. Moreover, the stacked design reduces potential leak sites, ensuring the system is both user-friendly and versatile with easily exchangeable stages.

Using the new SNaP mixer design, we demonstrate the ability to synthesize a wide range of PEG-PLA NPs and microparticles loaded with rubrene, a fluorophore, or cinnarizine, a weakly hydrophobic drug commonly prescribed to treat nausea and vomiting. To our knowledge, this is the first time that small NPs under 200 nm and MPs synthesized via SNaP have been reported. This SNaP set-up also enabled us to demonstrate for the first-time empirical evidence of the importance of the T_d in controlling the particle size. We introduced two straightforward methods for varying this T_d. Finally, we demonstrate for the first time, the successful encapsulation of drugs in NPs and MP with SNaP. Overall, we used 3D printing to prototype mixers for SNaP using a modular system that allowed us to rapidly screen process parameters. Using this 3D-printed SNaP mixing platform, we reproducibly synthesized polymeric particles for drug delivery and fluorescence imaging, with acute size control.

2. Results and Discussion

Herein, we describe the 3D printed mixer development process, including the major design considerations and the advantages of the resulting modular design. We then demonstrate NP and MP synthesis via SNaP, including the influence of T_d on particle size. Finally, we evaluate the 3D printing mixer reproducibility and mixer robustness.

2.1 Mixer Development via 3D-Printing

The original FNP mixer designs have traditionally been machined from blocks of Delrin or stainless steel.[35] This has several advantages, including organic solvent resistance and smooth inner channels for controlled fluid flow. However, this method of subtractive manufacturing poses several operational challenges. For the CIJ, the internal fluid channels must be drilled into the block of material,[23] creating drill entrance holes that are potential leak points during use. The significant time, cost, and knowledge required to machine CIJ and MIVM mixers means that production is often outsourced to third party manufacturers. This limits the ability to rapidly prototype new mixer designs. As a result, the original CIJ and MIVM designs for FNP have not been adapted to meet the needs of the newly developed SNaP process. Because SNaP requires two mixing steps, users have had to manually connect two mixers in series arrangement using external tubing, which significantly limits control over the T_d between core formation and NP stabilization, a critical process parameter.

As SNaP is a nascent approach to NP manufacturing, we were interested in exploring new design strategies for controlling the T_d and improving on the available mixing geometries. Because, prototyping a mixer requires flexibility of design, low cost of production, and rapid turnaround of production, we decided to use a 3D printing approach for mixer generation.[36] With recent advancements in 3D printing, namely major improvements in resolution and an expansion in printable material options, it is now possible to 3D print smooth narrow channels using organic solvent resistant materials. [36–38]

To 3D print the mixers, we chose to use the PolyJet printing technique. PolyJet printing is a layer-by-layer printing method, in which droplets of UV-curable resins are deposited followed by UV curing. [39–41] This method has sufficient resolution for millifluidic channels and creates smooth surfaces and water-tight surface contact between pieces. We evaluated the organic solvent resistance of several commercially available resins for PolyJet printing (data not shown). Veroclear, a proprietary resin described as a polymethyl methacrylate-like by Stratasys, was chosen for its resistance to organic solvents and translucency, allowing for ease of internal leak detection. Accordingly, we used the Stratasys Objet30 Pro printer to print the SNaP mixers.

To allow for translation from a 3D-printed design to a machined mixer in the future, we chose a stacked design similar to the original MIVM design. Each mixer component was an individual equidimensional stage - inlet, 1st mixer, 2nd mixer - which could by layered one on top of the other like a sandwich, enabling easy assembly and disassembly (Figure 1). Each stage was made to be interchangeable, allowing for rapid screening of process parameters, specifically mixing geometry and mixer-to-mixer channel length. When a new geometry or path length was desired, only the relevant stage needed to be printed and replaced. This method of piecewise prototyping expedited parameter screening and reduced production time, material consumption, and material costs. Mixers were printed, post-processed and assembled within one day, and tested over a similar timescale. As such, designing, manufacturing, and testing a new mixer was completed within 1–2 days.

Our final SNaP mixer designs are shown in Figure 1 along with technical drawings in the SI (Figure S2). Two designs are presented. The first design is for the formation of NPs, which connects two 4-inlet MIVMs in series. We termed this the 4–4 mixer configuration. In this mixer configuration, there are three stages, each with a different function: (1) a top stage with seven inlets, (2) a middle stage with the first 4-inlet MIVM and three channels to allow the passthrough of streams for the second mixer, and (3) a bottom stage with the second 4-inlet MIVM and an outlet. These stages were fitting using a Teflon O-ring, and then joined using shoulder bolts and nuts (SI Figure S1). The second design is for the formation of MPs. Here we departed from the traditional SNaP mixer geometry, which links a CIJ to an MIVM, and instead opted to use a two-inlet MIVM linked to a four-inlet MIMV. We termed this the 2–4 mixer configuration. The rationale for this change in geometry was that 3D-printing and machining MIVM geometries are technically easier than CIJ geometries, and that vortex mixing can accommodate asymmetrical stream flow rates. In the 2–4 mixer configuration, there are three stages, each with a different function: (1) a top stage with five inlets, (2) a middle stage with the 2-inlet MIVM and three channels to allow the passthrough of streams for the second mixer, and (3) a bottom stage with the 4-inlet MIVM and an outlet. These stages were fitting using a Teflon O-ring, and then joined using shoulder bolts and nuts (SI Figure S1). The use of shoulder bolts not only allowed for the precise alignment of the channels during assembly, but also allowed for an extremely effective seal to be formed when tightened. Moreover, this bolted assembly method also allowed for easy disassembly and fine-tuning within a relatively short time frame.

These designs have several advantages over initial SNaP setups. First, in both stacked designs, we replaced external tubing between the mixing steps with printed channels. Precise control over the T_d between mixing steps can be tuned by altering the fluid flow rate or by altering the channel path length from the first mixer outlet to the second mixer inlet. The latter was achieved by printing different versions of the second mixing stage (see SI Figure S5). Second, vortex mixing geometries have better mixing and can accommodate input streams of different flow rates, unlike CIJ mixing geometries. The use of vortex mixing enables the use of asymmetric stream flow rates and thus a larger range of solvent to antisolvent ratios.

Additionally, using a stage-based configuration allows for easy in situ modification of mixer geometry. Finally, using 3D printed materials significantly reduced the cost and timing of manufacturing. A single stainless steel MIVM currently costs $5000 to purchase, with a week of lead time before shipping. As two mixers for SNaP are needed, the base mixer cost is $10,000. A complete 3D printed SNaP mixer, however, costs approximately $250, with only 1–2 days required to fully process a mixer. A visual comparison between traditional SNaP setups and the modular 3D printed setup can be found in the SI (Figure S3).

2.2 Tunable Particle Synthesis Through Process Parameters and Mixer Design

In SNaP, the formation of the particle core and particle stabilization are decoupled into two sequential assembly steps. It has been hypothesized that the T_d between the two mixing stages can control the final particle size.[34] Increasing the T_d should increase particle size, due to the longer core-formation period prior to stabilization in the second mixing stage. The modified Smoluchowski's model for NP formation via diffusion-limited aggregation, shown in equation 1, describes the relationship between core-formation time and particle radius, in which R is particle radius, t is formation time, T is temperature, c_{core} is mass concentration of core material, υ is solution viscosity, ρ is core material mass density, and k_B is the Boltzmann constant. From equation 1, we see that longer formation times t should result in larger particle sizes.

$$R = \left(\frac{t k_B T c_{core}}{\pi \upsilon \rho_{core}}\right)^{\frac{1}{3}} \quad (1)$$

In this study, we demonstrated empirically for the first time that the T_d does influence the final particle size (Figure 2D and 4D) and can be used as an additional lever to control particle size. As will be described in detail, we achieved control over the T_d via two strategies with millisecond resolution: (1) changing the process parameters or (2) changing the mixer design. Importantly, we discovered that with the 2–4 mixer configuration, we could synthesize MPs, an entirely new size regime of polymeric particles, which cannot be achieved with traditional FNP. Using the 2–4 and 4–4 SNaP mixers, we synthesized particles ranging in size from 160 nm to 1.2 µm, greatly expanding what is achievable via FNP. For clarity, the following sections are organized by particle size regime - NP followed by MP SNaP synthesis.

2.3 SNaP for Nanoparticle Synthesis

For drug delivery and biomedical imaging applications, the NP size is a key physicochemical parameter.[42] We began by investigating how the T_d in the “4–4” SNaP mixers influenced the size of NPs. We define the T_d as the time it takes fluid from the center of the first mixing chamber to enter the second mixing chamber. The T_d can be calculated using equation 2 in which $\boldsymbol{V}$ is the volume of the channel leading from the first mixing geometry to the second, Q is the volumetric flow rate of each stream i entering the first mixing geometry. The total delay volume, V, is comprised of the first stage outlet channel, V’, and the volume of the second stage channel leading into the second mixer, which is calculated by multiplying the channel cross sectional area, a by the channel length, L. Volume and T_d calculations for each mixer configuration can be found in the SI.

$$T_d = \frac{V}{\sum Q_i} = \frac{V' + aL}{\sum Q_i} \tag{2}$$

From equation 2, we see that flowrate and T_d are inversely proportional. Thus flowrate, which is an easily tunable process parameter, can be used to control the T_d’s. As such, we chose to use flowrate to modulate the T_d’s in our NP syntheses. However, there are upper and lower bounds on the process flow rates and thus achievable T_d’s. At the upper bound, we are limited by the maximal backpressure that the syringe pumps can handle. For our pump and chosen 4–4 mixer SNaP geometry, the maximal flowrate attainable was 36 mL/min, which equates to a 5.8 ms T_d in our chosen set-up. At the lower bound, we are limited by the flow rate requirements to achieve micromixing. Micromixing is necessary to ensure homogeneous supersaturation of components prior to assembly events.[22] In MIVM mixers, micromixing is achieved at Reynolds numbers (Re) above 1600.[22] The Re for MIVM mixers are calculated using equation 3 in which U_i is the velocity of each incoming stream i to the mixing geometry, and υ_i is the kinematic viscosity of the stream. Re calculation details can be found in the SI. In our chosen 4–4 SNaP geometry, the minimal flow rate to achieve micromixing is 2.1 mL/min (Re = 1600), which results in a T_d of 100.4 ms. For NP formation, much shorter T_d’s are desired. Thus, the flowrate range tested was between 20 mL/min (Re= 15559, 23740; $T_d = 10.5\ \text{ms}$) and 36 mL/min (Re = 28006, 42733; $T_d = 5.8\ \text{ms}$).

$$Re = \sum_{i = 1, N} \frac{U_i}{v_i} D \tag{3}$$

With these bounds in mind, we used a model fluorescent NP system to evaluate the role of T_d on NP size. The system was composed of rubrene and PLA homopolymer for the core and PEG-PLA as the stabilizer. As shown schematically in Figure 2A, the core components dissolved in THF were mixed against three aqueous anti-solvent streams to initiate the core formation. The core was stabilized in the second mixing step with a stream of PEG-PLA in THF and two anti-solvent streams. We kept the stream concentrations constant across all experiments and varied the flow rates to change in the inter-mixer T_d. NPs were successfully synthesized at all flow rates tested. A representative intensity weighted DLS nanoparticle size trace and corresponding TEM image of spherical nanoparticles made with the shortest delay time are shown in Figure 2 B and C.

We observed that as the T_d increased, the size of the rubrene-loaded NPs increased. For the shortest T_d tested (5.8 ms), the NP size was 198 ± 2 nm, while for the longest T_d tested (10.5 ms), the NP size increased to 233 ± 6 nm. In Figure 2D, particle size as a function of T_d is plotted on a log-log scale. From the slope, we see that the particle size scales as 1/3, in agreement with Smoluchowski's model of diffusion-limited nanoparticle growth (equation 1). This supports the hypotheses that (1) delaying the addition of the stabilizer (increasing T_d) allows the cores to continue growing, resulting in larger nanoparticles, and that (2) the T_d is a useful process parameter for controlling particles size. Experimental conditions and results are summarized in Table 1.

To expand beyond our model fluorescent system, we applied the SNaP process to form drug-loaded NPs. To our knowledge, this is the first time that drug-loaded NPs formed by SNaP have been reported. We encapsulated cinnarizine, an antihistamine, in PEG-PLA NPs. Cinnarizine is a weakly hydrophobic drug that has a logP of 5.7; it has been previously demonstrated that its encapsulation into NPs via FNP requires the hydrophobic ion pairing (HIP) solubility engineering approach.[31] HIP involves the addition of an ionizable hydrophobic counterion species, which complexes with an ionizable API to drive its nucleation and growth.[43] Here, we added a hydrophobic counter ion, xinafoic acid with a logP of 3.2, to the core organic stream to form a hydrophobic complex with cinnarizine to promote its encapsulation. Using the 4–4 mixing configuration, and with a T_d of 5.8 ms, we successfully formed cinnarizine-loaded PEG-PLA NPs with a size of 169 nm. The NPs had a drug loading of 4.0 ± 0.2 wt.% and an encapsulation efficiency of 80 ± 3%. Size distribution and encapsulation results are summarized in Figure 3. To our knowledge, this is the first time that small NPs under 200 nm synthesized via SNaP have been reported. These results demonstrate that SNaP is an effective tool for the formation of fluorophore and drug-loaded polymeric NPs for bioapplications.

2.4 SNaP for Microparticle Synthesis

Polymeric MPs are also of interest for drug delivery and biomedical imaging applications. While FNP has seen significant success making NPs, it has never successfully been used to create polymeric particles greater than 400 nm. We wanted to test if decoupling the core formation and stabilization, allowing cores to grow for longer, would enable us to access the MP size regime.

We began by investigating how the inter-mixer T_d in the 2–4 SNaP mixers influenced the size of particles. Instead of using the flowrate to control the T_d as previously described, we generated 2–4 mixers with varying channel lengths between the two mixing steps. As shown in equation 2, the T_d is directly proportional to the channel length. We printed three second mixer stages of varying channel lengths to achieve three different T_d's, while maintaining constant stream flow rates. The secondary stage mixer designs can be found in the SI (Figure S5).

To evaluate the role of T_d on particle size, we used the same model fluorescent system with rubrene, PLA and PEG-PLA. At a fixed inlet flow rate of 30 mL/min (Re = 13370), the resulting T_d's were calculated to be 13.5 ms, 18.9 ms, and 22.5 ms for each respective mixer configuration. As shown schematically in Figure 4A, the core components dissolved in THF were mixed against an aqueous anti-solvent stream to initiate the core formation. The core was stabilized in the second mixing step with a stream of PEG-PLA in THF and two anti-solvent streams. Excitingly, at these longer delay times, MPs were successfully synthesized. A representative intensity weighted DLS nanoparticle size trace and corresponding confocal image of spherical MPs made at a T_d of 13.5 ms are shown in Figure 4 B and C.

We again observed that the size of rubrene-loaded MPs increased with increasing T_d. For the shortest delay channel length tested ($T_d = 3.5\ \mathrm{ms}$), the MP size was 0.96 ± 0.02 µm, while for the longest delay channel length tested ($T_d = 22.5\ \mathrm{ms}$), the particle size increased to 1.19 ± 0.11 µm. In Figure 4D, particle size as a function of T_d is plotted on a log-log scale. From the slope, we see that the particle size scales as 1/3, in agreement with Smoluchowski's model of diffusion-limited particle growth. These results support the hypothesis that T_d is a key process parameter in controlling MP size. To explore this further, an even shorter T_d was achieved in the 2–4 configuration by increasing the flow rate on the shortest delay channel length configuration from 30 to 60 mL/min (the same strategy used in the previous 4–4 investigation). This gave a T_d of 6.8 ms and resulted in even smaller MPs (0.76 ± 0.07 µm) which also showed experimental agreement with the Smoluchowski model. This agreement with the model, despite using different methods of tuning Td, further reinforces the parameter as key in controlling particle size. Experimental conditions and results are summarized in Table 2.

To expand beyond our model fluorescent system, we applied the SNaP process to form drug-loaded MPs again using cinnarizine and the HIP formulation approach with xinafoic acid. To our knowledge, this is the first time that drug-loaded MPs formed by SNaP have been reported. With the 2–4 mixer configuration with the 13.5 ms delay time, we successfully synthesized 1.1 µm cinnarizine-loaded MPs. The MPs had a 4.2 ± 0.1 wt.% cinnarizine drug loading with an encapsulation efficiency of 83.8 ± 2.7%. Size distribution and encapsulation results are summarized in Figure 5. By loading active pharmaceutical ingredients into MPs via SNaP, we demonstrate the potential of this process to generate polymeric particles across both the nanometer and micron size range.

2.5 3D-Printed Mixer Reproducibility

3D printed prototyping of the SNaP mixers enabled rapid screening of process parameters due to the ease of printing and re-printing mixer stages. During the mixer design optimization, dozens of mixers were printed and reprinted. For this reason, it was imperative to verify that the 3D printing process itself would not introduce stage variability or imperfections that could affect the particle synthesis process. In such a case, findings from one mixer iteration could not confidently be used to design the next. Furthermore, it is imperative to show that the 3D printing process as a whole is reliable enough to be used for millifluidic mixers. Therefore, we investigated the reproducibility across mixer prints. Three identical mixers were printed, and an identical MP formulation was synthesized on each mixer three times. The size and PDI results for each mixer can be seen in Figure 6. MP size averaged 1.08 μm across all three mixers. T-tests confirmed no statistical significance between mixer and both size and PDI ($P = 0.81$ and 0.63, respectively). The insensitivity of particle size to mixer replicate suggests that the described 3D printing protocol did not invoke random variation that could affect particle assembly. This supports that this 3D printed prototyping approach showed major advantages with regard to speed and cost, with no discernable loss of reproducibility or reliability. It also shows that mixing geometries and configurations can be re-printed for rapid hardware replacements or mass production.

2.6 3D-Printed Mixer Prototype Failure-Point Testing

To evaluate the robustness of the 3D printed mixers, the lifespan of 3D printed prototype mixers was determined. Lifespan was measured as number of particle syntheses a mixer could tolerate until a significant change in particle size was observed. An identical MP formulation was produced on a mixer repeatedly, measuring size with each synthesis. In between syntheses, blank runs were conducted using only solvents, as dissolved polymer and fluorophore were presumed to have no degenerative effects on the 3D printed channels. Every fifth run, a fluorescent particle synthesis was performed, with solvent streams containing dissolved PEG-PLA, PLA, and rubrene. The results are summarized in Figure 7A, where hydrodynamic diameter as a function of the syntheses number is plotted for three identical 3D printed mixers. Mixers were robust, showing no statistically significant change in MP size and polydispersity over 21 uses.

The typography of 3D printed mixer channels was investigated before and after 21 uses using SEM. We imaged the mixing chamber and channels of a newly printed mixer and a mixer after 21 uses. As presented in Figure 7B and 7C, both the newly printed mixer and heavily used mixers show similar surface topography in all channels and mixing geometries, with a slightly smoother surface for the used mixer. No change in any of the dimensions of the mixer was observed after use.

These results show that 3D printing provided a tool for rapidly prototyping SNaP mixers that enabled us to achieve a controlled and reproducible synthesis of polymeric particles. This mixer geometry can achieve production levels of up to 15L/h of particle solution at the highest flow rates. This production level could be further increased by scaling up mixers' dimensions and by using pressure pumps. The 3D printed SNaP mixers are ideal for lab-scale production and for rapid prototyping of the mixer design. Nevertheless, these 3D printed mixers have some limitations, such as the risk of channel swelling that may occur with continuous organic solvent exposure. While not used in this manuscript, we have observed that the mixers are incompatible with dimethyl sulfoxide (DMSO). We observed degradation of the mixers when used with DMSO underflow. Therefore, for continuous applications, it is advisable to transition to machined SNaP mixers with the same optimized geometries made from more durable materials such as Delrin or stainless steel.

3. Conclusions

In this work, we developed a robust, low-cost, millifluidic SNaP mixing platform and demonstrate for the first time the size-tunable assembly of polymeric NPs and MPs and successful drug encapsulation via the SNaP process. Instead of the traditional method of connecting machined mixers in a series arrangement, we developed a stacked mixer design, which eliminated the need for external tubing and gave greater control over mixer geometries, including the inter-mixer T_d. Using our stacked mixer design, we unlocked a new size scale that is unattainable with the traditional FNP – the micron size scale. This is the first time that MPs formed via SNaP have been reported. Additionally, we demonstrated the ability to make small NPs via SNaP. Thus, the stacked mixer design enables the synthesis of particles from the nano to the micron range, broadening the applicability of SNaP. Importantly, we demonstrated for the first time that the T_d is a key process parameter and that it can be used as a lever to tune particle size. We demonstrated that the nanoparticle size scales with T_d in agreement with Smoluchowski's model of diffusion-limited particle growth. In addition to fluorophores, we encapsulated the drug cinnarizine in both NPs and MPs with high encapsulation efficiency. This is the first time that drug-loaded NPs and MPs synthesized via SNaP have been reported. Finally, we demonstrated the reproducibility of the 3D printed mixers by generating nearly identical particle sizes across several prints. The robustness of these mixers was also tested, and no degradation from use was observed. Together, these findings demonstrate the potential of the SNaP continuous flow manufacturing process to make tunable particles for a broad range of bioimaging and drug delivery applications.

Our new SNaP mixer set-up enables precise control of core growth, opening new pathways for investigating the assembly mechanics of polymeric particles with more complex core structures. Our future work will focus on investigating the impact of the core composition, aggregation kinetics, and the component addition order on encapsulation efficiency and final particle properties, potentially unlocking new particle architectures.

4. Experimental Section/Methods

Materials:

Tetrahydrofuran (THF, HPLC grade), acetonitrile (ACN, HPLC grade), and cinnarizine (98.0%) were purchased from Fisher Scientific (USA). Dimethyl sulfoxide (DMSO, HPLC grade), polylactic acid homopolymer (PLA, 10–18 kD), rubrene (98%), and 1 μm & 3 μm polystyrene microparticle standards were purchased from Sigma-Aldrich (USA). 2-hydroxy-1-napthaldehyde (xinafoic acid) was purchased from Thermo Fisher Scientific. Poly(lactic acid) – poly(ethylene glycol) (PEG-PLA, 5k-5k) was donated by Evonik Inc (USA). Ultra-pure water (18.2 MΩ·cm) was generated by a MilliporeSigma Milli-Q Water Purification System (USA). Veroclear resin was purchased from Stratasys (USA).

For assembly of 3D printed mixers, Tefzel tubing for inlets (0.040” ID) and outlets (1/16” ID), female LuerTight syringe fitting systems (1/16” OD), VacuTight Fittings (1/4–28 – 1/8), and flangeless male nut fittings (1/4–28, 1/16”) were purchased from Idex Health and Science (USA). O-rings (75 viton, 1.5 × 35 mm) and heated inserts for inlets were purchased from McMaster Carr (USA).

3D Printing and Assembly of Mixers:

The mixer stages were designed using Autodesk Fusion360 and printed on a Stratasys sObjet30 Pro using Veroclear following the manufacturer’s instructions. Post-printing, the support material was removed under running water using prying tools. The threaded heat inserts were added to the mixer using an arbor press. Once all eight input inserts and a final output insert were set, compressed air was blown through the inlets to remove all processing residue. O-rings were then set into the two mixing stages. All the mixer stages were then assembled and secured using shoulder bolts, washers, and nuts.

Two mixer designs were developed to enable the synthesis of particles over a wide range of particle sizes. The first design, the 2–4 mixing configuration, is comprised of two mixing stages. The first stage is a two-inlet vortex mixer, followed by a second stage, a 4-inlet vortex mixer. The second design, the 4–4 mixing configuration, is comprised of two mixing stages. The first stage is a four-inlet vortex mixer, followed by a second stage that is a 4-inlet vortex mixer. The mixer stage designs and dimensions can be found in the SI (Figure S2).

Nanoparticle Formation:

NPs were synthesized via Sequential Nanoprecipitation (SNaP) using the described 3D-printed 4–4 mixer configuration. Before and following the NP synthesis, all mixer streams were first flushed with 3 mL of THF (30 mL/min) followed by 3 mL of water (30 mL/min).

For the synthesis of rubrene NPs, a THF stream with dissolved PLA (9.7 mg/mL) and rubrene (0.3 mg/mL) was first mixed against 3 streams of ultra-pure water in the first mixing stage. The output of this first mixing stage became an inlet stream for the second mixing stage, where it was mixed against one THF stream containing dissolved PEG-PLA (20 mg/mL), and two streams of ultra-pure water. To achieve the desired T_d, the flow rate for each stream was set at either 36 mL/min, 30 mL/min, or 20 mL/min using a syringe pump (PHD Ultra Syringe pump, Harvard Apparatus).

For the synthesis of cinnarizine-loaded NPs, we first prepared a THF stream with dissolved PLA (7.66 mg/mL), cinnarizine (1.5 mg/mL), and xinafoic acid (hydrophobic counterion, 0.84 mg/mL). This organic stream was mixed against 3 streams of ultra-pure water in the first mixing stage. The output of this first mixing stage became an inlet stream for the second mixing stage, where it was mixed against a THF stream containing dissolved PEG-PLA (40 mg/mL), and two streams of ultra-pure water. The flow rate for each stream was 36 mL/min. NPs were purified via dialysis to remove unencapsulated drug and counter ion. Using regenerated cellulose dialysis tubing (6–8kDa MWCO, Repligen), particles were dialyzed against a large excess of ultrapure water for six hours, changing the water every hour.

To account for start-up volumes, NP samples were collected only after the flow had fully developed to simulate a continuous flow system. This solution was collected in a 4°C MillQ water quenching bath to reduce the final organic concentration to 10 vol% THF.

Microparticle Formation:

Particles were synthesized via Sequential Nanoprecipitation (SNaP) using the described 3D-printed 2–4 mixer configuration. Prior to particle synthesis, mixers were prepped by first flowing 3 mL of THF through all streams (30 mL/min), followed by 3 mL of water through all streams (30 mL/min).

For the synthesis of rubrene MPs, a THF stream containing dissolved PLA (39.2 mg/mL) and rubrene (0.8 mg/mL) was first mixed against an equal volume of ultra-pure water in the first mixing stage. The output of this first mixing stage became an inlet stream for the second mixing stage, where it was mixed against one THF stream containing dissolved PEG-PLA (40 mg/mL), and two streams of ultra-pure water. The flow rate per stream was set at 30 mL/min. To tune the T_d, three different second mixing stages were used (Figure S5).

For the formation of drug-loaded MPs, we first prepared the core organic stream that is composed of THF containing dissolved PLA (30.6 mg/mL), cinnarizine (4 mg/mL), xinafoic acid (hydrophobic counterion, 4.6 mg/mL) and rubrene (0.8 mg/mL). This organic stream was mixed against an equal volume of ultra-pure water in the first mixing stage. The output of this first mixing stage became an inlet stream for the second mixing stage, where it was mixed against one THF stream containing dissolved PEG-PLA (40 mg/mL), and two streams of ultra-pure water. Again, the flow rate per stream was set at 30 mL/min. MPs were purified via dialysis to remove unencapsulated drug and counter ion. Using regenerated cellulose dialysis tubing (6–8kDa MWCO, Repligen), particles were dialyzed against a large excess of ultrapure water for six hours, changing the water every hour.

To account for start-up volumes, MP samples were collected only after the flow had fully developed to simulate a continuous flow system. This solution was collected in a 4°C MilliQ water quenching bath to reduce the final organic concentration to 10 vol% THF.

Particle Characterization:

Dynamic light scattering (DLS) size measurements were performed using a ZetaSizer Nano ZS (Malvern Instruments, USA) after nanoprecipitation using a detection angle of 173° at 25°C. NP samples were diluted 10x in ultrapure water, while MP samples were diluted 5x in ultrapure water. Intensity-average diameter as presented by the Malvern deconvolution software general purpose mode is reported, with standard deviation reported across 3 measurements. Because DLS is not commonly used to analyze particles in the micron region, DLS accuracy was evaluated in the 1–3 μm diameter range using polymeric MP standards with known size distributions (see SI, Figure S9).

Total particle solids concentration was determined using a thermogravimetric analyzer (TGA, Discovery 550 TGA, TA Instruments, USA). Particles in solution were heated under nitrogen from 25°C to 105°C at a rate of 10°C/min and held at 105°C for 20 minutes to ensure complete water evaporation. Total particle concentration was determined by dividing the final solid weight by the starting sample volume.

Cinnarizine concentration was determined via High Performance Liquid Chromatography on a Thermofisher Vanquish Core HPLC equipped with an autosampler, quarternary pump, UV detector, and column (Hypersil Gold C18, 3um particle size, 175 Å pore size, 4.6 mm x 150 mm, 30°C). HPLC samples were prepared by diluting NP solutions into acetonitrile to a final 60 vol% acetonitrile. The following isocratic mobile phase was used: 60 vol% acetonitrile with 0.1 vol% trifluoracetic acid and 40 vol% water with 0.1 vol% trifluoracetic acid, with a total flow rate of 0.5 mL/min for 20 minutes. Detection was performed at 224 nm.

NPs were visualized via transmission electron microscopy (TEM) using a FEI Talos L120C microscope. To prepare the TEM sample, a 0.5–1.0 mg/mL NP solution was deposited onto the formvar on carbon, 300 mesh Cu TEM grid (TedPella), and allowed to dry. Negative staining was done with 2% phosphotungstic acid, pH 6. All TEM imaging was performed using the instrument and facilities at NYU Langone Microscopy Laboratory (RRID: SCR_017934).

MPs were visualized via scanning electron microscopy (SEM). To prepare the samples, MPs were concentrated via centrifugation for 5 minutes at 100 rcf. Concentrated particles were plated dropwise onto silicon wafers. After drying, samples were sputtered with a thin conductive platinum layer (3 nm) using an EMS 150R ES sputter coater. They were then imaged using the SE2 detector on the Zeiss Merlin FE-SEM microscope, operating at 1 kV EHT and 110 pA.

Failure-Point Testing for 3D-Printed Mixers:

Mixer longevity was determined with a failure-point testing protocol. A 2–4 mixer was printed and used to synthesize several MP batches. Mixer failure was defined as the number of microparticle syntheses reached when output MPs showed a significant deviation in size distribution from the first batch synthesized.

The previously described rubrene MP formulation and process was used for this test. For every microparticle synthesis, four blank runs were performed to reduce the use of materials. Blank runs consisted of identical volumes and flowrates of the solvent and antisolvent streams, but without polymer or rubrene. The mixer was used 21 times, with MP syntheses taking place on the 1st, 6th, 11th, 16th, and 21st run. To achieve statistical significance, the failure point testing was performed on three identical mixers. Particle size was characterized via DLS immediately post-synthesis.

SEM Imaging of 3D Printed Mixers Channels:

SEM was used to image the surface topography of 3D printed mixer channels. Both a newly printed 2–4 mixer and a used 2–4 mixer (from the failure point test) were bisected to fit on the imaging stage. The mixer pieces were sputtered with a thin conductive gold layer (3–5 nm) utilizing an EMS 150R ES sputter coater. Subsequently, they were imaged using the SE2 detector on the Zeiss Merlin FE-SEM microscope, operating at 5 kV EHT and 110 pA.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

Thomas Belinky, Nouha El Amri, and Parker K. Lewis contributed equally to this work. This work was funded by grants from the National Institutes of Health (R01 DE02995101S1) and from the Department of Defense (W81XWH2210238). PKL would like to acknowledge financial support from the Eli Pearce Graduate Fellowship program. NEA would like to acknowledge financial support from the NYU Provost Postdoctoral Fellowship Program. The authors acknowledge the use of the shared instrumentation facilities provided through the Materials Research Science and Engineering center (MSREC) and MRI programs of the National Science Foundation under awards numbers DMR-1420073 and DMR-0923251. The authors also acknowledge the use of the NYU Langone Microscopy Laboratory (RRID: SCR_017934) and the Cancer Center Support Grant P30CA016087. The NYU Tandon Makerspace is acknowledged for providing access to the Stratasys Objet30 Pro and for 3D printing support. We thank Prof. Weiqiang Chen for access to the Nikon Eclipse Ti2 Confocal Microscope. We also thank Prof. Trinanjana Mandal for her assistance with SEM imaging.

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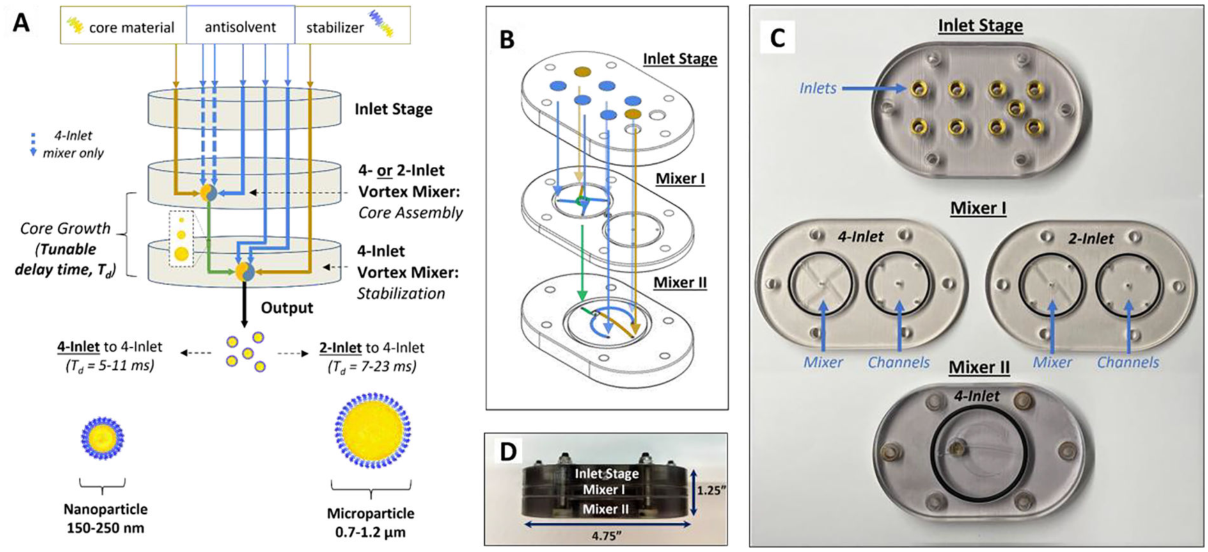

Figure 1.

A) A stacked mixer schematic, with an input stage for syringe connections (top), which connects immediately to the first mixing stage (middle). The first mixing stage is interchangeable, with either a 2-inlet or a 4-inlet mixer option depending on the desired particle size regime (dotted antisolvent streams only present in the 4-inlet mixer). This stage also contains pass-through for streams used in the second mixing step. All the streams mix in the second mixing stage (bottom) and exit the device. B) Flow path illustration of 4–4 mixer configuration. C) Photo of disassembled 3D printed mixer stages: inlet stage (top), first mixing stage; 4 -inlet (middle left) or 2-inlet (middle right) mixing geometry, and second mixing stage (bottom) with 4-inlet mixing geometry. D) Side view of assembled 3D printed mixer.

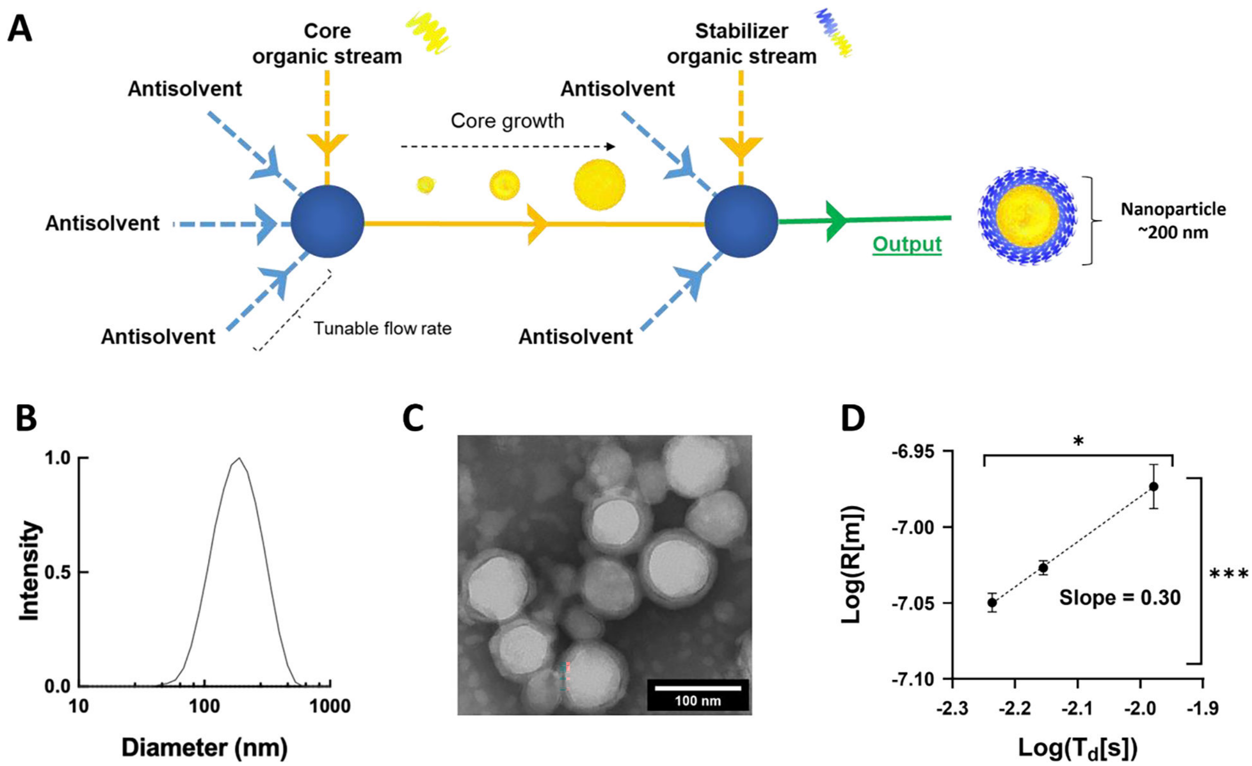

Figure 2.

A) Millifluidic stacked mixer flow path and SNaP schematic B) Representative nanoparticle diameter distribution from dynamic light scattering C) Representative TEM image of SNaP nanoparticles D) Log of Nanoparticle size vs log of inter-mixer delay time, (n=3). *Significant correlation by Pearson correlation analysis, $p < 0.05$ ***Significantly non-zero slope from nonlinear regression, $p < 0.0001$.

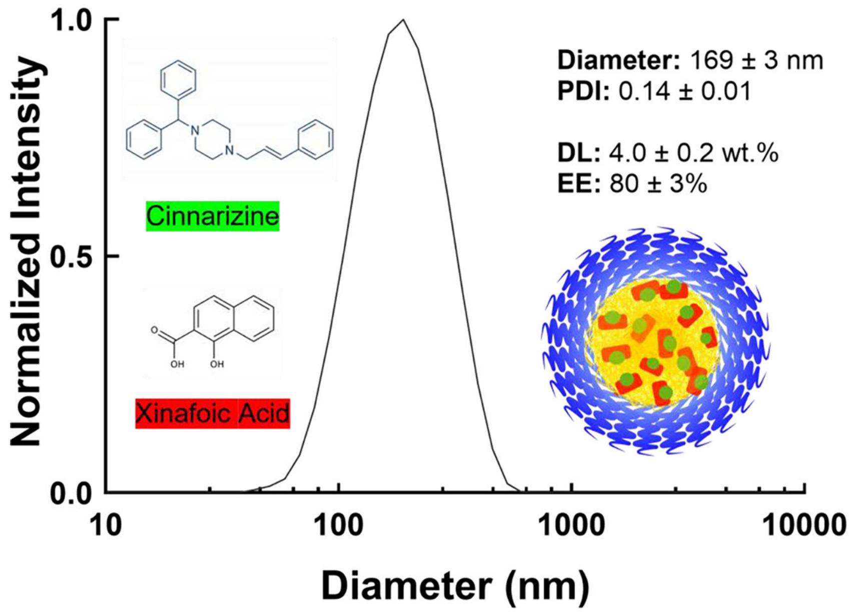

Figure 3.
Representative DLS size distribution of cinnarizine-loaded nanoparticles, showing average hydrodynamic diameter, PDI, drug loading (4%) and encapsulation efficiency (80%), (n=3).

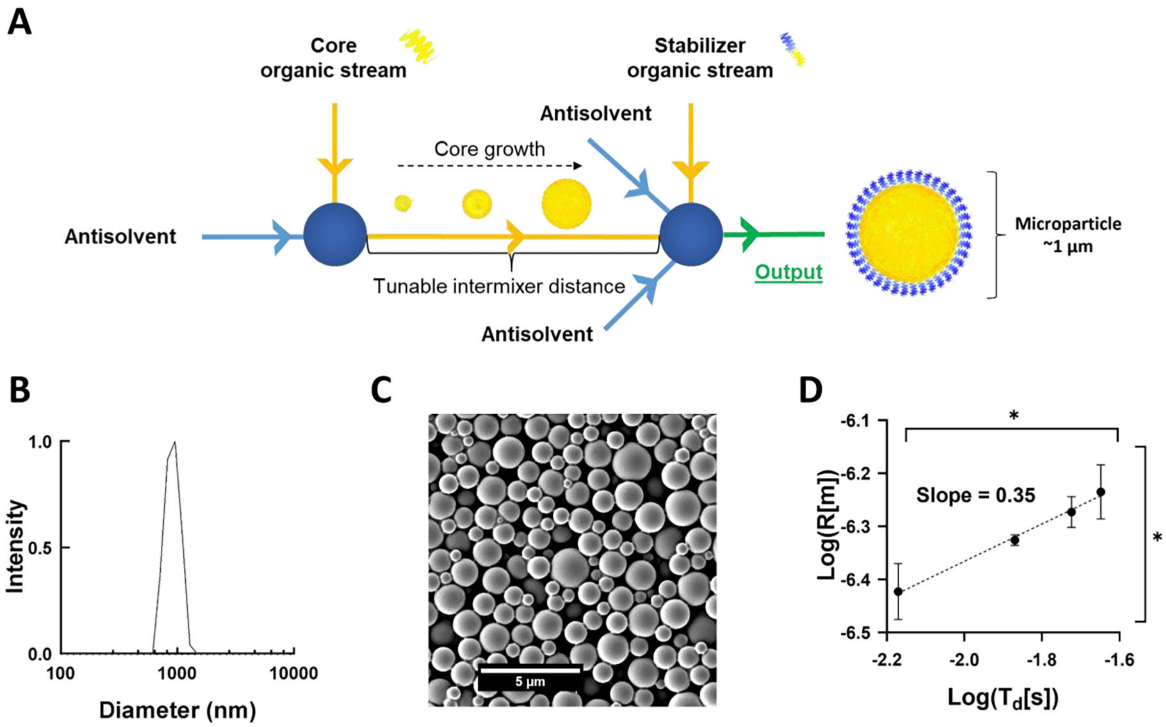

Figure 4.

A) Millifluidic stacked mixer flow path and SNaP schematic B) Representative microparticle diameter distribution from dynamic light scattering C) Representative SEM microparticle image D) Log of microparticle size vs log of inter-mixer delay time, (n=3). *Significant correlation by Pearson correlation analysis, $p < 0.05$, *Significantly non-zero slope from nonlinear regression, $p < 0.05$.

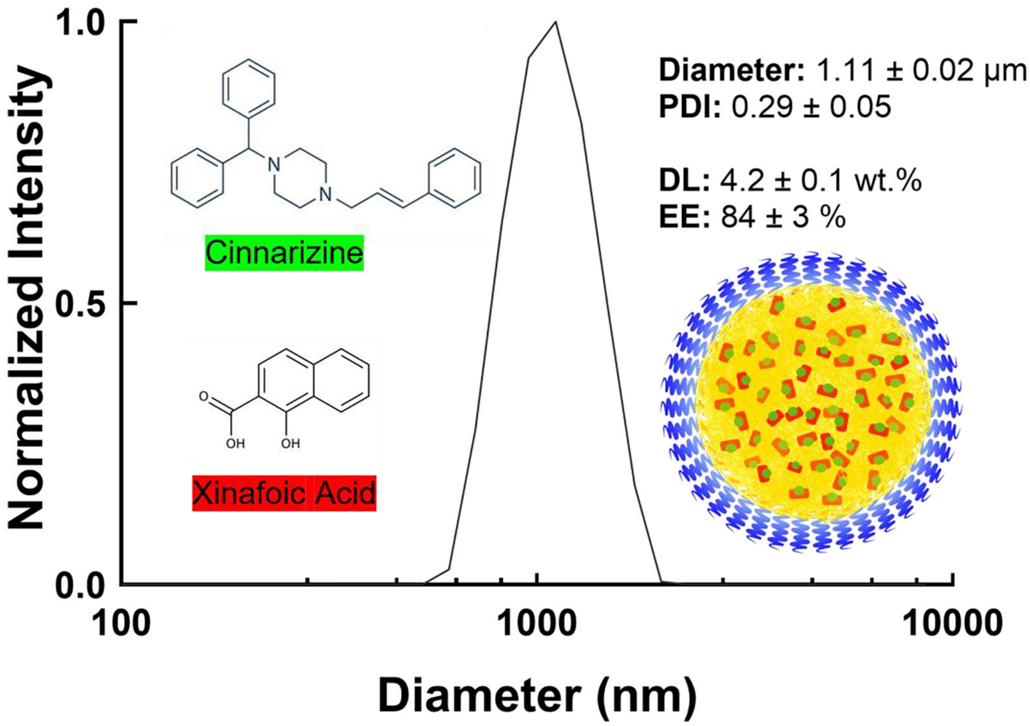

Figure 5.

Representative DLS size distribution of cinnarizine-loaded microparticles, showing average hydrodynamic diameter, PDI, drug loading (wt.%) and encapsulation efficiency (%) (n=3).

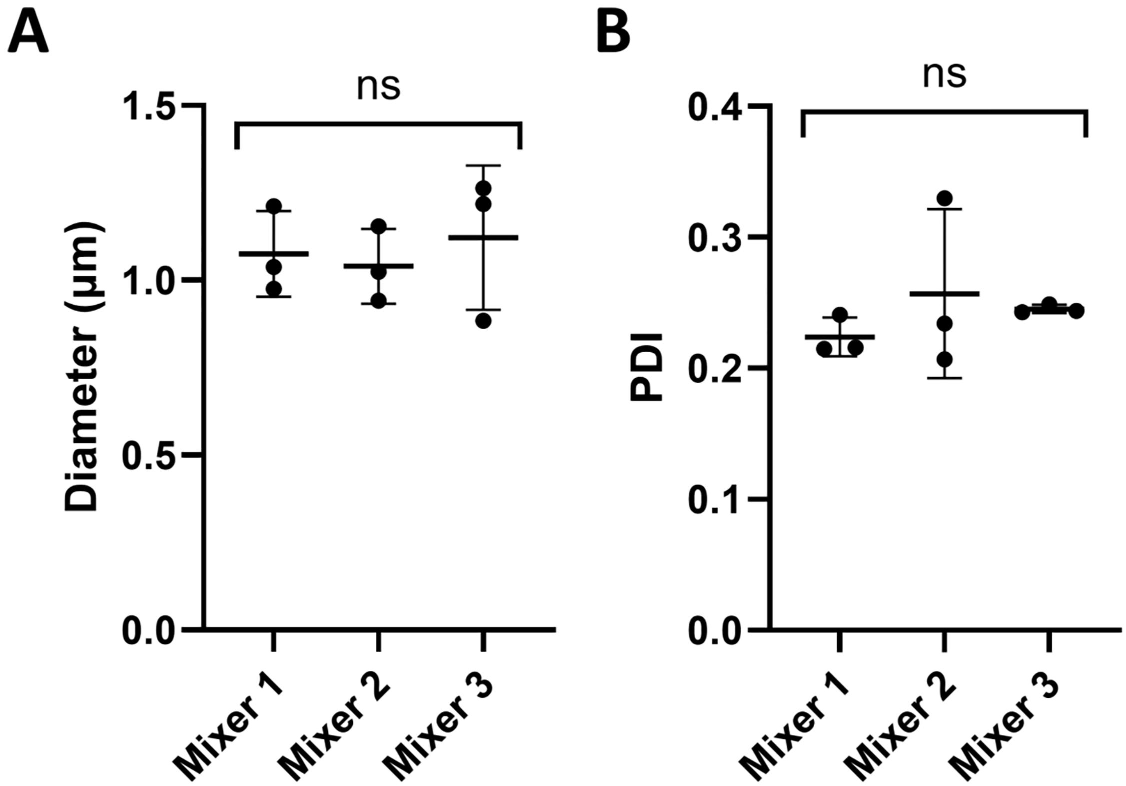

Figure 6.

Synthesis reproducibility of A) particle size and B) polydispersity index (PDI) of identical formulations performed on three identical mixers (n=3 per mixer). Similarity suggests 3D printing protocol results in mixers with consistently identical particle assembly mechanics. ns = no significant difference in size, Brown-Forsythe & Welch's ANOVA ($p > 0.5$ for both tests on diameter and PDI).

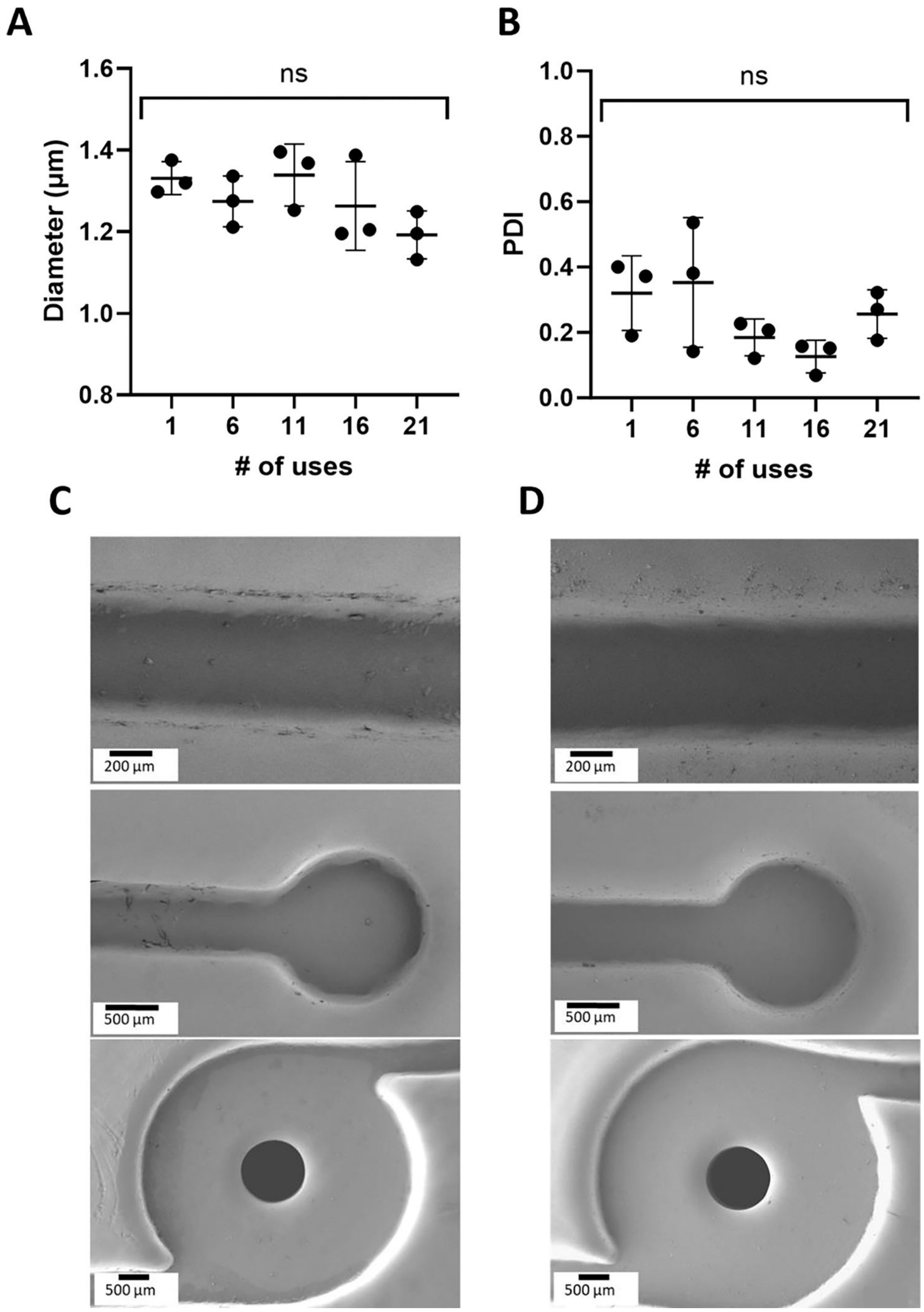

Figure 7.

A) Microparticle hydrodynamic size and B) polydispersity index (PDI) from DLS over repeated syntheses across three mixers. Each data point is from one mixer. ns = non-significant difference in size over 21 uses, 2way ANOVA, $p > 0.05$. C) SEM image of inlet channels (top, middle) and mixing geometry (bottom) of an unused 3D printed mixer. D) SEM image of inlet channels (top, middle) and mixing geometry (bottom) of a 3D printed mixer after 21 particle syntheses.

Table 1.

SNaP parameters for nanoparticle size tuning via flow rate.

Mixer Design[a]	L[b] [mm]	V[c] [m^3]	Q[d] [mL/min]	T_d[e] [ms]	Re_1[f]	Re_2[g]	Diameter[h] [nm]
4-4	9.525	1.40E-08	**36**	5.8	28006	42733	198 ± 2
4-4	9.525	1.40E-08	**30**	7.0	23338	35611	208 ± 2
4-4	9.525	1.40E-08	**20**	10.5	15559	23740	233 ± 6

[a.] Mixer geometry can be found in the SI

[b.] Length of channel between mixing chambers

[c.] Volume between mixing chambers

[d.] Inlet stream flowrates

[e.] Delay time

[f.] Reynolds number for first mixing step

[g.] Reynolds number for second mixing step

[h.] Nanoparticle diameter.

Table 2.

SNaP parameters for microparticle size tuning via channel length.

Mixer Design[a]	L[b] [mm]	V[c] [m^3]	Q[d] [mL/min]	T_d[e] [ms]	Re_1[f]	Re_2[g]	Diameter[h] [μm]
2-4 A’	**9.53**	1.35E-08	**60**	6.8	26740	51812	0.76 ± 0.07
2-4 A	**9.53**	1.35E-08	30	13.5	13370	25906	0.96 ± 0.02
2-4 B	**19.05**	1.89E-08	30	18.9	13370	25906	1.08 ± 0.06
2-4 C	**25.40**	2.25E-08	30	22.5	13370	25906	1.19 ± 0.11

a. Mixer geometry can be found in the SI

b. Length of channel between mixing chambers

c. Volume between mixing chambers

d. Inlet stream flowrates

e. Delay time

f. Reynolds number for first mixing step

g. Reynolds number for second mixing step

h. Microparticle diameter.